Immersive Air Quality Explorer(IAQE)

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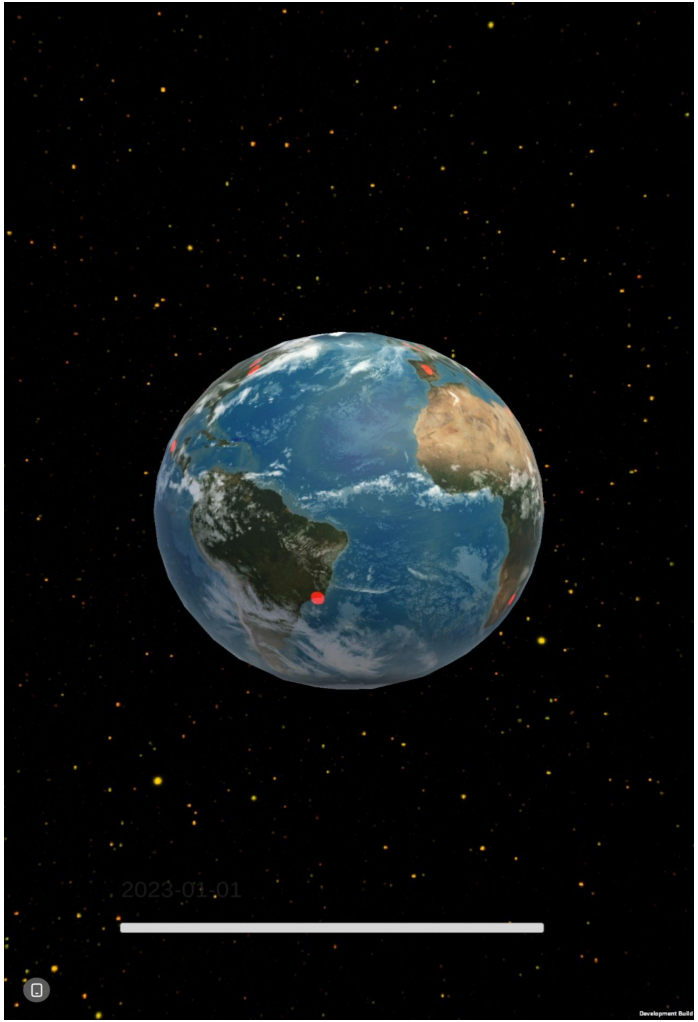
Why Immersive Air Quality Visualization?

- Global AQI data is complex + multi-dimensional (location, time, multiple pollutants).
- Traditional 2D views hide spatial relationships and make comparisons harder.
- AR + 3D turns air quality into something you can *walk around*, explore, and remember.
- Goal: help users discover hotspots, compare cities, and see changes over time interactively.

Immersive Air Quality Explorer (IAQE) — What It Does

- Global overview: 3D globe with city markers (visual hotspot scanning).
- City drill-down: tap/select a city to reveal AQI + pollutants (PM2.5, PM10, NO_2, SO_2, CO, O_3).
- Time exploration: timeline slider changes date → values/visuals update.
- PM10 analytics: a pie chart shows the overall distribution of PM10 levels across all cities in the dataset.
- AR interactions: explore using tap + device movement; designed for ARCore devices.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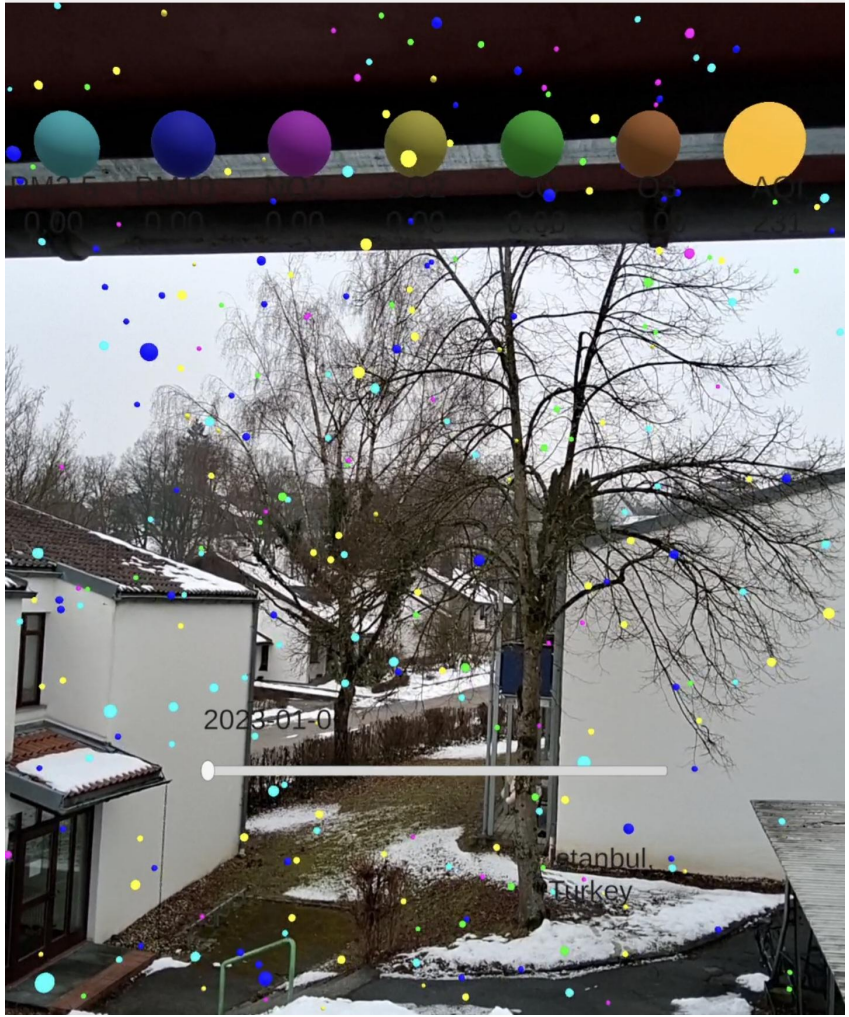
Development Build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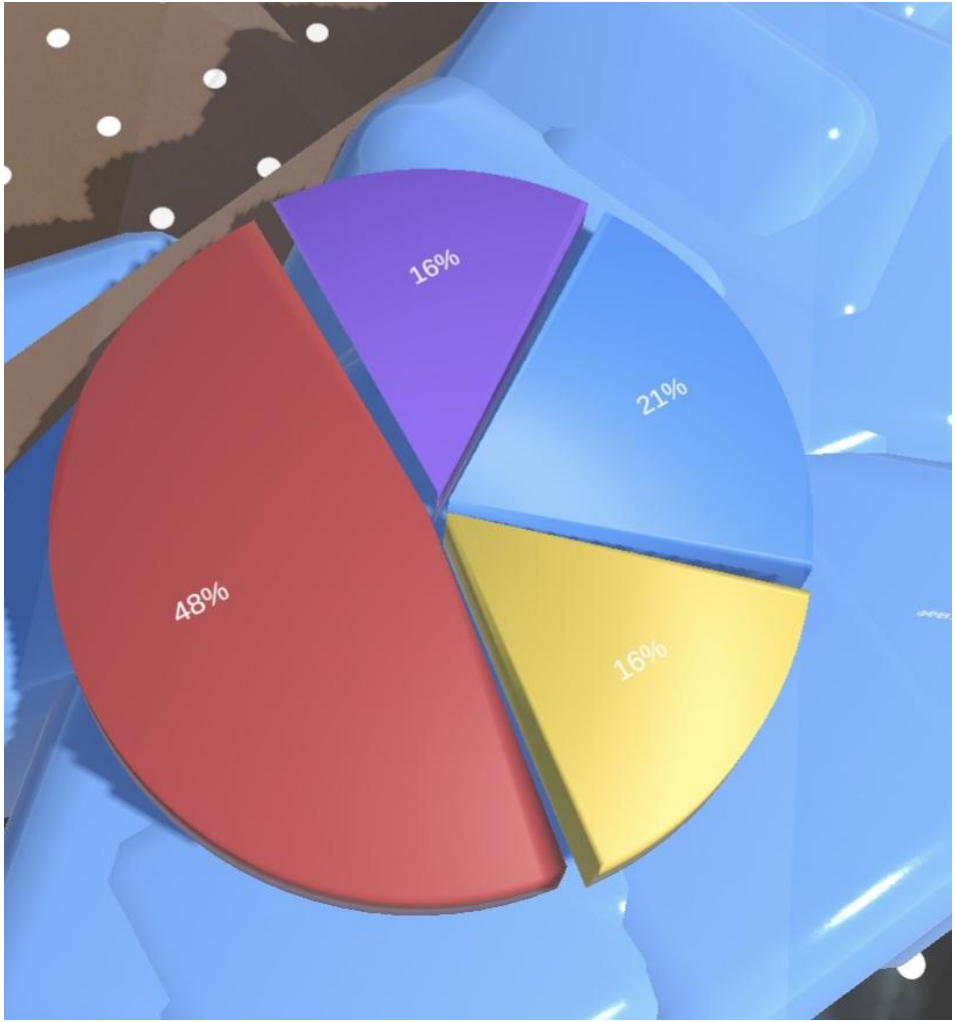
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Data → Visual Mapping

- Dataset: global air quality CSV with city/country + lat/long + date + AQI + pollutants.
- Position: latitude/longitude → correct geospatial placement (globe).
- Size: bubble size represents AQI severity.
- Color: color represents pollution severity category (quick scanning).
- Details-on-demand: selecting a city shows pollutant breakdown.
- PM10 impact : Pie chart shows the proportion of records/cities where PM10 is the dominant pollutant, highlighting how often PM10 drives overall air-quality impact.

Architecture and Implementation

Data Layer: CSV import → parsing → date indexing (fast lookup by day).

Visualization Layer (Unity): globe renderer + city bubble spawner + UI overlays.

Interaction Layer (AR): tap/select, rotate/zoom gestures, timeline slider; built/tested with Unity + ARCore devices.

Limitations

- Visual clutter at global scale (many cities close together).
- Selection accuracy/occlusion challenges in AR.
- Dataset coverage/quality varies by location/date.

Thank You